

ZERO-TO-JOB-READY

DATA ANALYST 2026

M08 - STATISTICS

Variation -> samples -> uncertainty -> tests -> effect -> causality -> experiments

QUESTION -> ESTIMATE -> UNCERTAINTY -> BUSINESS THRESHOLD -> LIMITATION

One number never decides a business experiment.

This book is your teacher

- Statistics here is analyst reasoning, not advanced mathematics. Every formula is tied to a business question and plain-language interpretation.
- Do not memorize p-values/CI rules as slogans. State the estimate, units, uncertainty, assumptions, business threshold and limitation separately.
- Videos make abstract ideas visible. The workbook and simulation notebook give changed practice; protected review opens only after a genuine attempt.
- Large n improves precision but cannot repair selection bias. Statistical significance does not guarantee practical importance. Correlation does not establish causation.
- Experiment decisions use design quality + effect size + uncertainty/p-value + business threshold + guardrail—not one number.

Normal statistics loop

1. Frame population/question
2. Define metric/units
3. Watch exact visual if useful
4. Follow calculation once
5. Do guided sheet
6. Close support + changed task
7. Validate independently
8. Interpret + limitation + evidence

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Module prerequisite and tool map

0. Prerequisite

M00-M07 analyst reasoning: data grain, quality, grouped summaries, percent/ratio thinking, charts and evidence. No advanced mathematics is assumed. Calculator/Excel/Python are tools; interpretation remains the learner's job.

Tools / files

- C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx - direct STAT1-STAT9 sheets + M-STAT.
- C1_M08_STATISTICS_VISUAL_SIMULATION_NOTEBOOK_RC1.ipynb - visual companion after matching lesson.
- C1_M08_STATISTICS_PROTECTED_PRACTICE_REVIEW_RC1.pdf - post-attempt only.
- Gate A/B/C workbooks and separate protected reviews.
- Calculator/Excel/Python are allowed when assigned; you must still explain the metric and decision in ordinary language.

Workbook architecture repaired

The inherited practice workbook merged/shifted STAT4-STAT9. RC1 gives each canonical lesson its own guided + changed practice sheet, while preserving useful experiment data.

This prevents correlation/A-B tasks from appearing under the wrong lesson IDs.

Exact practice, project and Gate route

- STAT1 descriptive -> STAT2 distributions/z -> STAT3 sampling/bias -> STAT4 sampling distributions/SE/CLT -> STAT5 CI -> STAT6 p-values -> STAT7 effect/business importance -> STAT8 correlation/causal limits -> STAT9 A/B design.
- Every lesson: learner book -> mapped visual if useful -> guided workbook -> changed task -> independent validation -> explain-back -> append evidence.
- Complete M-STAT on Experiment_Data. Predefine metric, worthwhile effect and guardrail before interpreting the result.
- Attempt Gate A independently. A clean pass ends M08.
- Gate B/C are fresh remediation full retests only after prior failure + protected review + targeted repair + explicit assignment.

Critical failures

p-value treated as probability H_0 is true; $p > 0.05$ treated as proof of no effect; narrow CI treated as proof of no bias; statistical significance used as business value; correlation called causal; or guardrail/design/data-quality failure ignored can block a pass.

STAT1 - Descriptive statistics: center, spread and outliers

Practice: STAT1_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Choose and interpret mean, median, percentiles, IQR, range and sample standard deviation without reducing a dataset to one misleading number.

3. What you already need / prerequisite

No prior statistics formula is assumed.

4. Picture it in your head

Imagine describing a classroom. The mean is everyone's score averaged; the median is the middle person; SD/IQR describe how spread out the class is. One unusual score may pull the mean without representing the typical student.

5. New words before we use them

Mean = arithmetic average. Median = middle ordered value. Percentile = relative position. Range = max-min. IQR = $Q3 - Q1$. Sample SD = spread around sample mean. Outlier fence = investigation rule, not deletion command.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - Calculating the Mean, Variance and Standard Deviation, Clearly Explained!!! | Full focused video. Why watch: Visual explanation of center/variance/SD before the course adds median/IQR/outlier investigation. Internal fallback: STAT1 learner book + workbook fully teach center/spread/outlier rules.

[Open visual source](#)

STAT1 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Mean uses every value and is sensitive to extremes. Median is resistant to extreme observations. Standard deviation describes spread around the mean; IQR describes the middle 50%. An outlier flag is a reason to investigate, not a deletion command. Worked example: Delivery minutes [20, 22, 24, 25, 150]. The mean is pulled upward by 150 while the median stays near the typical routine delivery. For a highly right-skewed metric, report median plus percentiles/IQR alongside the mean rather than claiming one number is the whole story. Guided action: Use the practice sheet to calculate center/spread on delivery times and identify possible outliers using the $1.5 \times \text{IQR}$ rule.

8. What you should see / be able to say

Recompute one statistic independently, verify the row count, and explain whether any flagged outlier is a data error, rare valid case, or unknown.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Reporting only the mean for skewed data; automatically deleting an extreme value because it is inconvenient.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT1 - Do alone, validate and master

12. Now you do a small version alone

Analyze a changed support-resolution dataset. Decide which center/spread summary is most defensible and explain why.

13. Validate your result

Recompute one statistic independently, verify the row count, and explain whether any flagged outlier is a data error, rare valid case, or unknown.

14. Teach it back in your own words

Without the book open, explain when mean vs median is more representative and why an outlier rule does not automatically justify deletion.

15. Protected review + changed fresh retry

Save the STAT1 workbook/simulation attempt first. If repair is needed, open the protected STAT1 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT2 - Distributions, percentiles, normal models and z-scores

Practice: STAT2_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Read distribution shape, use percentiles and z-scores, and state when a normal model is or is not a reasonable approximation.

3. What you already need / prerequisite

Prerequisite: STAT1 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

A distribution is a landscape of values. Percentiles tell where a value sits in that landscape. A z-score measures distance from the mean in SD units only when mean/SD and model assumptions are useful.

5. New words before we use them

Distribution = pattern/spread of values. Normal model = symmetric bell-shaped probability model. z-score = $(\text{value} - \text{mean}) / \text{SD}$. P90 = 90th percentile. Skew = asymmetric tail. Unusual = rare relative to model, not automatically erroneous.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - The Normal Distribution, Clearly Explained!!! | Full focused video. Why watch: See the normal model visually before deciding whether z-score/normal language fits a business metric. Internal fallback: STAT2 learner book + workbook teach distribution shape, percentiles and z-score limits.

[Open visual source](#)

STAT2 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A distribution shows how values are spread. Percentiles express relative position. A z-score is $(\text{value} - \text{mean}) / \text{standard deviation}$. Normal models are useful only when the shape/assumptions are a reasonable approximation. Worked example: If mean handle time is 12 minutes and SD is 3, an 18-minute call has $z = +2$. It is two standard deviations above the mean. That makes it unusual relative to the model, not automatically invalid. Guided action: Compare a near-symmetric service-time sample with a skewed purchase-value sample. Compute selected percentiles and z-scores.

8. What you should see / be able to say

Check shape visually and compare mean/median before using normal-model language.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Assuming every business metric is normal; calling a high z-score automatically wrong.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT2 - Do alone, validate and master

12. Now you do a small version alone

Given a changed dataset, choose between percentile-based language and z-score/normal-model language and defend the choice.

13. Validate your result

Check shape visually and compare mean/median before using normal-model language.

14. Teach it back in your own words

Explain what a z-score says and what it does not say.

15. Protected review + changed fresh retry

Save the STAT2 workbook/simulation attempt first. If repair is needed, open the protected STAT2 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT3 - Population, samples and sampling bias

Practice: STAT3_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Define the target population, sample, sampling frame and main bias risks before generalizing.

3. What you already need / prerequisite

Prerequisite: STAT2 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

Population is everyone you want to talk about; sampling frame is the actual list you can reach; sample is who you observed. If the frame misses a group, more responses from the same frame do not magically include them.

5. New words before we use them

Target population = group you want to infer about. Sample = observed units. Sampling frame = units eligible to be selected. Sampling error = chance sample variation. Sampling bias = systematic selection distortion. Nonresponse = selected units do not respond.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - Sampling from a Distribution, Clearly Explained!!! | Full focused video. Why watch: Makes random sampling variation visible before the course distinguishes chance error from systematic sampling bias. Internal fallback: STAT3 learner book + sampling-design sheet fully teach population/frame/bias/generalization.

[Open visual source](#)

STAT3 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Random sampling error is chance variation between samples. Sampling bias is systematic and does not disappear just because the sample is large. Selection, nonresponse and survivorship can distort conclusions. Worked example: A survey sent only to recent support users can overrepresent frustrated customers. Ten thousand such responses can be less representative than a smaller probability sample of all active customers. Guided action: Classify four sampling designs by population, frame, likely bias and repair.

8. What you should see / be able to say

State who could never be selected under your design and whether that limits the target population.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Generalizing beyond the sampled population; believing sample size repairs selection bias.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT3 - Do alone, validate and master

12. Now you do a small version alone

Design a sampling plan for a changed customer-satisfaction question with a constrained budget.

13. Validate your result

State who could never be selected under your design and whether that limits the target population.

14. Teach it back in your own words

Explain the difference between random sampling error and sampling bias.

15. Protected review + changed fresh retry

Save the STAT3 workbook/simulation attempt first. If repair is needed, open the protected STAT3 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT4 - Sampling distributions, standard error and the CLT

Practice: STAT4_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Explain why estimates vary from sample to sample and how sample size affects standard error and precision.

3. What you already need / prerequisite

Prerequisite: STAT3 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

Take many samples from the same population. Each sample mean moves a little. Those sample means form their own distribution. SE describes that estimator movement; it usually shrinks as n grows.

5. New words before we use them

Statistic = number calculated from sample. Sampling distribution = distribution of a statistic across repeated samples. Standard error (SE) = SD of estimator/sampling distribution. CLT = sample means often approach normal under broad conditions as n grows.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - The Central Limit Theorem, Clearly Explained!!! | Full focused video. Why watch: Visualize repeated sample means and why their spread changes with n . Internal fallback: STAT4 learner book + visual simulation notebook teach SD vs SE and CLT.

[Open visual source](#)

STAT4 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A statistic such as a sample mean varies across repeated samples. Standard error describes variability of an estimator. For a mean, SE is commonly approximated by sample SD / $\sqrt{n}$. The CLT explains why sample means often become approximately normal under broad conditions as n grows. Worked example: If two surveys have the same SD but one has 9 times as many observations, its standard error is roughly one-third as large because $\sqrt{9}=3$. More precision does not repair a biased sample. Guided action: Use the workbook and simulation notebook to compare repeated sample means for $n=10$, 40 and 160.

8. What you should see / be able to say

Check that the observed spread of repeated sample means decreases with n while the underlying population spread stays similar.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Confusing SD with SE; confusing precision with representativeness.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT4 - Do alone, validate and master

12. Now you do a small version alone

Explain how a changed sample-size decision affects precision without claiming it fixes bias.

13. Validate your result

Check that the observed spread of repeated sample means decreases with n while the underlying population spread stays similar.

14. Teach it back in your own words

Explain SD vs SE and why CLT is about the sampling distribution, not magically normalizing the raw data.

15. Protected review + changed fresh retry

Save the STAT4 workbook/simulation attempt first. If repair is needed, open the protected STAT4 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT5 - Confidence intervals and uncertainty

Practice: STAT5_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Compute and interpret approximate confidence intervals while separating precision from bias.

3. What you already need / prerequisite

Prerequisite: STAT4 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

A confidence interval is a ruler around an estimate built by a repeated procedure. A narrower ruler means more precision under assumptions; it cannot reveal selection bias by itself.

5. New words before we use them

Point estimate = one estimated value. Confidence interval (CI) = interval from a procedure. Confidence level = long-run capture rate under assumptions. Precision = interval narrowness. Bias = systematic shift from target; narrow does not imply unbiased.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - Confidence Intervals, Clearly Explained!!! | Full focused video. Why watch: Visual CI intuition before the course focuses on repeated-procedure interpretation, precision and bias. Internal fallback: STAT5 learner book + workbook + simulation notebook fully teach the CI interpretation contract.

[Open visual source](#)

STAT5 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

A 95% confidence procedure is designed to capture the true parameter in about 95% of repeated samples under its assumptions. A narrower interval means greater precision, not automatically less bias. Always report the estimate itself. Worked example: Estimated conversion lift +0.5 percentage points with 95% CI [-0.2pp, +1.2pp] does not exclude zero at the 95% level. Positive business-relevant effects may still be plausible. Guided action: Calculate approximate mean CIs for $n=100$ and $n=900$ with the same mean and SD.

8. What you should see / be able to say

Verify the SE arithmetic and state at least one assumption/limitation.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Saying a completed CI has a 95% probability of containing a fixed true parameter; treating a narrow biased interval as trustworthy.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT5 - Do alone, validate and master

12. Now you do a small version alone

Interpret a changed interval for a difference in business terms, including effects still compatible with the data.

13. Validate your result

Verify the SE arithmetic and state at least one assumption/limitation.

14. Teach it back in your own words

Explain what a 95% CI procedure means and why interval width is not a bias test.

15. Protected review + changed fresh retry

Save the STAT5 workbook/simulation attempt first. If repair is needed, open the protected STAT5 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT6 - Hypothesis testing, null hypothesis and p-values

Practice: STAT6_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Interpret a p-value under a pre-declared decision rule without treating it as a truth probability.

3. What you already need / prerequisite

Prerequisite: STAT5 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

Start with a baseline/no-effect story H_0 . Ask: if that story were true, how surprising would data this extreme be? The p-value answers that conditional question—not the probability that H_0 itself is true.

5. New words before we use them

Null hypothesis H_0 = baseline/no-effect model. Alternative = competing effect model. Alpha = predeclared significance cutoff. p-value = tail probability of data this/extreme under H_0 . Reject/fail to reject = decision language, not truth proof.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - Hypothesis Testing and The Null Hypothesis, Clearly Explained!!! | 0:22-13:44. Why watch: See null/reject/fail-to-reject logic before applying the analyst decision rule. Internal fallback: STAT6 learner book + workbook teach two-proportion test and safe p-value language. Optional companion: StatQuest - p-values: What they are and how to interpret them | 0:17-9:47.

[Open visual source](#)

[Open optional companion](#)

STAT6 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

The null hypothesis is the baseline/no-effect model. A p-value asks how surprising the observed data or more extreme data would be if the null were true. $p < \alpha$ is statistically significant under that rule; failing to reject H_0 is not proving H_0 . Worked example: If $p=0.08$ at $\alpha=0.05$, write: the data do not provide enough evidence to reject the null at 0.05. Do not write: there is no effect. Guided action: Use the practice A/B summary to compute/interpret a two-proportion z-test at analyst depth.

8. What you should see / be able to say

State the null, alpha, estimate, p-value and decision separately.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Saying $p=0.03$ means a 3% probability the null is true; saying $p>0.05$ proves no effect.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT6 - Do alone, validate and master

12. Now you do a small version alone

Interpret a changed p-value/CI pair without overstating certainty.

13. Validate your result

State the null, alpha, estimate, p-value and decision separately.

14. Teach it back in your own words

Explain p-value, significance and non-significance in ordinary language without the two common false interpretations.

15. Protected review + changed fresh retry

Save the STAT6 workbook/simulation attempt first. If repair is needed, open the protected STAT6 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT7 - Effect size, practical significance and decision thresholds

Practice: STAT7_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Separate evidence of a difference from whether the difference is large enough to matter.

3. What you already need / prerequisite

Prerequisite: STAT6 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

Statistical evidence asks whether a difference is distinguishable from chance under assumptions. Business significance asks whether the size is worth money/risk/effort. These are separate decision axes.

5. New words before we use them

Absolute lift = treatment-control in percentage points. Relative lift = absolute lift/control. Minimum worthwhile effect = business threshold. Guardrail = metric not allowed to worsen beyond limit. Practical significance = effect matters operationally/economically.

6. Watch / visual source

OPTIONAL_VISUAL_COMPANION. StatQuest - p-values: What they are and how to interpret them | Ending emphasis: p-values do not measure effect size. Why watch: Useful bridge from significance to effect size/business thresholds. Internal fallback: STAT7 learner book + workbook are the canonical teacher.

[Open visual source](#)

STAT7 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Absolute lift = treatment rate - control rate. Relative lift = absolute lift / control rate. Tiny effects can be statistically significant in huge samples; valuable effects can be uncertain in small samples. Define a minimum worthwhile effect and guardrails before seeing results when possible. Worked example: Control conversion 5.0%, treatment 5.4%: absolute lift +0.4 percentage points; relative lift 8%. Whether +0.4pp matters depends on economics, risk and costs. Guided action: Calculate absolute/relative lift and compare it with a stated minimum worthwhile effect and a guardrail threshold.

8. What you should see / be able to say

State estimate, uncertainty, business threshold and guardrail separately.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Calling every statistically significant result important; ignoring a damaged guardrail metric.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT7 - Do alone, validate and master

12. Now you do a small version alone

Make a decision for a changed experiment where statistical significance and business importance point in different directions.

13. Validate your result

State estimate, uncertainty, business threshold and guardrail separately.

14. Teach it back in your own words

Explain why a p-value cannot tell you whether an effect is large enough to matter.

15. Protected review + changed fresh retry

Save the STAT7 workbook/simulation attempt first. If repair is needed, open the protected STAT7 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT8 - Correlation, confounding and causal claims

Practice: STAT8_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Quantify linear association while respecting the limits of observational evidence.

3. What you already need / prerequisite

Prerequisite: STAT7 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

Two lines can move together because X causes Y, Y causes X, or Z moves both. Correlation measures linear association; the causal story needs stronger design/evidence.

5. New words before we use them

Pearson r = linear correlation from -1 to +1. Confounder = third variable related to both X and Y. Reverse causality = Y influences X. Association = variables vary together. Randomization = chance treatment assignment to balance confounders on average.

6. Watch / visual source

RECOMMENDED_VISUAL. StatQuest - Pearson's Correlation, Clearly Explained!!! | 0:55-17:31. Why watch: Visual interpretation of Pearson correlation before the course adds confounding/reverse-causality limits. Internal fallback: STAT8 learner book + workbook + simulation teach association vs causation. Optional companion: Khan Academy - Correlation and causality | Full focused video.

[Open visual source](#)

[Open optional companion](#)

STAT8 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Pearson correlation ranges from -1 to +1 for linear association. High correlation can be caused by confounding, reverse causality, selection effects or a real direct effect. Randomized experiments are usually stronger for causal inference. Worked example: Advertising spend and sales can be correlated because larger stores both spend more and sell more. The association can be real while the causal story is wrong. Guided action: Compute Pearson correlation for store ad spend vs sales, then list plausible confounders and a stronger design.

8. What you should see / be able to say

Inspect the scatter pattern and confirm that one or two extreme points are not driving the whole association.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Calling correlation causal; ignoring common causes and reverse causality.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT8 - Do alone, validate and master

12. Now you do a small version alone

Interpret a changed observational correlation and write one defensible claim plus one prohibited causal claim.

13. Validate your result

Inspect the scatter pattern and confirm that one or two extreme points are not driving the whole association.

14. Teach it back in your own words

Explain association vs causation and name at least three reasons a correlation can mislead a causal claim.

15. Protected review + changed fresh retry

Save the STAT8 workbook/simulation attempt first. If repair is needed, open the protected STAT8 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation.
PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

STAT9 - A/B tests and experimental thinking

Practice: STAT9_... sheet in C1_M08_STATISTICS_CANONICAL_PRACTICE_WORKBOOK_RC1.xlsx

1. Why this lesson matters

Business data varies. A good analyst must summarize that variation without turning a useful statistic into a false certainty. This lesson teaches one reasoning tool and its interpretation boundary.

2. What you are learning

Design and interpret a basic randomized business experiment with primary metric, guardrails, quality checks and a pre-declared decision rule.

3. What you already need / prerequisite

Prerequisite: STAT8 concepts plus ordinary percentage/ratio reasoning. The new statistical idea is taught from first principles.

4. Picture it in your head

An A/B test is a contract written before results: who is randomized, what success means, what could be harmed, how long you run, what effect matters, and what decision rule you will follow.

5. New words before we use them

Control = baseline experience. Treatment = changed experience. Unit of randomization = entity assigned. Primary metric = predeclared main outcome. Guardrail = protected harm metric. Sample ratio = observed allocation vs plan. Peeking = repeated unplanned stopping checks.

6. Watch / visual source

OPTIONAL_VISUAL_COMPANION. Khan Academy - Correlation and causality | Use as a bridge: why randomization strengthens the causal argument. Why watch: Contrast observational association with randomized experimental design. Internal fallback: STAT9 learner book + experiment workbook teach the complete A/B design discipline.

[Open visual source](#)

STAT9 - Follow with me once

7. Follow with me once - fully worked teacher demonstration

Define treatment/control and unit of randomization. Predefine primary metric, guardrails, duration/stopping rule and decision threshold. Randomization helps balance confounders on average. Check sample-ratio problems, data loss and implementation failures. Avoid opportunistic peeking and post-hoc metric switching. Worked example: A clean question: among eligible visitors during a planned 14-day period, does a simplified signup page increase signup conversion without materially increasing support contacts? Guided action: Audit an A/B-test plan, then analyze the integrated experiment dataset.

8. What you should see / be able to say

Confirm group counts, metric definitions, missing/duplicate IDs and guardrail integrity before interpreting the result.

9. Why it worked

The statistic is connected to the correct population/sample/metric and units, and the interpretation stays inside its assumptions. A second calculation, simulation, visual or quality check provides independent evidence instead of trusting one formula cell.

10. Common beginner mistakes

Changing metrics after results; repeatedly peeking and stopping at $p < 0.05$; presenting exploratory segments as confirmed effects.

11. If you get stuck - recovery ladder

1) Restate the business question and unit. 2) Write the sample/population and metric. 3) Check counts/data quality. 4) Recompute the simplest component. 5) Use the visual simulation for intuition if assigned. 6) Open protected review only after saving the attempt.

STAT9 - Do alone, validate and master

12. Now you do a small version alone

Write a pre-analysis plan for a changed experiment with unit, primary metric, guardrail, duration and decision threshold.

13. Validate your result

Confirm group counts, metric definitions, missing/duplicate IDs and guardrail integrity before interpreting the result.

14. Teach it back in your own words

Defend an experiment decision using design quality, effect size, uncertainty, p-value and guardrails - not one number.

15. Protected review + changed fresh retry

Save the STAT9 workbook/simulation attempt first. If repair is needed, open the protected STAT9 review, diagnose only the weak concept, close it, then complete the changed task without answer material. Append the retry.

16. Evidence / mastery + exact next route

Save calculation, units, sample/population or design assumptions, independent check, plain-language interpretation and limitation. PASS -> next STAT lesson. PARTIAL/FAIL -> targeted repair and changed retry before moving on.

M-STAT - Integrated experiment decision project

1. Why this project matters

Real experiment decisions combine data quality, rates, effect size, uncertainty, statistical evidence, business thresholds and guardrails. The project proves you can integrate them without letting one attractive number dominate the decision.

2. What you are building

A complete treatment-vs-control decision using Experiment_Data: data-quality checks, conversion/support rates, absolute/relative lift, approximate 95% CI, two-sided z/p-value, worthwhile-effect comparison, guardrail check, decision and limitations.

3. Prerequisite

STAT1-STAT9 fresh evidence. Do not change the primary metric, treatment labels, threshold or guardrail after seeing the result.

4. Picture it in your head

Experiment contract -> quality -> estimate -> uncertainty/test -> business threshold -> guardrail -> decision -> limitation. A statistically interesting result can still be a no-launch if the guardrail fails.

5. New words before we use them

Absolute lift = treatment-control pp. Relative lift = lift/control %. Two-proportion z test = analyst-depth test for two rates. CI for rate difference = uncertainty around lift. Decision contract = predeclared rule combining evidence/value/guardrail.

6. Watch / visual source

No new required video. Reuse only a named weak STAT lesson. Project mastery comes from your calculations, decision logic and defense.

M-STAT - Worked micro-example

7. Follow with me once - fully worked teacher demonstration

Before using Experiment_Data, use a tiny thought experiment: control $100/1000 = 10\%$; treatment $110/1000 = 11\%$. Absolute lift = +1pp; relative lift = +10%. If minimum worthwhile effect = +1.5pp, the observed effect misses the business threshold even before considering uncertainty. This separates 'effect exists?' from 'large enough to matter?'.

8. What you should see / be able to say

Percentage points compare rates directly; relative lift scales the difference by the control rate. A business threshold is not a p-value cutoff. Statistical evidence and practical importance answer different questions.

9. Why it worked

The example fixes the denominator and units before testing. That prevents common mistakes such as calling +1pp a 1% relative lift or treating $p < 0.05$ as the definition of business success.

10. Common beginner mistakes

Change metric after results; ignore sample-ratio/duplicate/missing checks; report relative lift without absolute pp; say CI including zero means no effect; launch on significance despite failing a guardrail.

11. If you get stuck - recovery ladder

Freeze the experiment contract. Recompute group counts and rates manually. Check units. Separate estimate, CI, p-value, business threshold and guardrail into five lines before writing a decision.

M-STAT - Full independent build and validation

12. Now you do the project

- State target population, unit, control/treatment, primary conversion metric, support-contact guardrail, $\alpha=0.05$ and minimum worthwhile effect before interpreting.
- Validate group counts, duplicate unit IDs, missing/invalid group/converted/support labels.
- Compute control/treatment conversion and absolute/relative lift.
- Compute support-contact rates and pp change.
- Compute approximate 95% CI for conversion-rate difference and a two-sided two-proportion z/p-value.
- Keep the estimate, CI, p-value, worthwhile-effect threshold and guardrail on separate lines.
- Make launch / do-not-launch / continue-test decision and explain each input.
- Create one useful chart/table; do not imply causality beyond randomized assignment/design integrity.
- State assumptions, implementation/data-quality risks and at least two limitations.
- Save evidence, then close protected material and use a fresh Gate only if assigned.

13. Validate

Reconcile group counts, conversion rates, lift units, support rates, CI sign, p-value decision, business threshold and guardrail. If any quality/design check fails, do not let a clean p-value override the problem.

M-STAT - Explain, evidence and project rubric

14. Teach it back in your own words

Give a two-minute defense: experiment question -> target population/unit -> quality -> control/treatment rates -> absolute/relative effect -> CI -> p-value/alpha -> worthwhile threshold -> guardrail -> decision -> limitation. Never say the p-value is $P(H_0 \text{ true})$.

15. Protected review + changed fresh retry

Save M-STAT first. If one concept fails, open only the relevant lesson/project review. Close it, repair the concept, then use changed Gate data only if the Gate state explicitly assigns it.

16. Evidence / mastery + exact next route

Critical pass: correct denominators/units; design/quality checked; effect size + uncertainty + p-value separated; business threshold predeclared and evaluated; guardrail evaluated; no false CI/p-value/causal statements; decision follows combined evidence; limitations stated; learner can defend without review. PASS -> Gate A. PARTIAL/FAIL -> targeted repair + changed project evidence before Gate A.

End-of-module mastery checklist

- STAT1: Choose/interpret center and spread; investigate outliers rather than auto-delete.
- STAT2: Read shape/percentiles/z-scores and state when normal-model language is weak.
- STAT3: Define population/sample/frame and distinguish random sampling error from systematic bias.
- STAT4: Explain SD vs SE, sampling distribution and why larger n improves precision, not representativeness.
- STAT5: Compute/interpret approximate CI and separate precision from bias.
- STAT6: State H_0 /alpha/p-value/decision without false probability claims.
- STAT7: Report absolute/relative effect and compare with worthwhile threshold/guardrail.
- STAT8: Quantify association while naming confounding/reverse-causality/causal boundary.
- STAT9: Predeclare A/B design, unit, metric, guardrail, duration, quality checks and decision rule.
- M-STAT: Defend an experiment decision using design + effect + uncertainty + p-value + business threshold + guardrail.

Gate readiness

Attempt Gate A only after M-STAT evidence is saved and false p-value/CI/causal interpretations are absent.

A clean Gate A pass ends M08; B/C are remediation full retests only.

Revision / statistics quick reference

Mean / median	Mean uses every value; median resists extremes. Report shape/spread too.
IQR	Q3-Q1; 1.5×IQR fence flags values to investigate.
z-score	Distance from mean in SD units; unusual does not mean invalid.
Sampling bias	Systematic frame/selection problem; larger n cannot repair it.
SE	Estimator variability; for mean, often $s/\sqrt{n}$. Not raw-data SD.
95% CI	Repeated procedure designed for ~95% coverage under assumptions.
p-value	$P(\text{data this/extreme} \mid H_0)$, not $P(H_0 \mid \text{data})$.
Effect size	Magnitude; report pp and relative % with business threshold.
Correlation	Linear association, not causation.
A/B decision	Design + quality + effect + uncertainty/p + threshold + guardrail.

Next route: M-STAT -> Gate A -> M09 Integrated Business Analysis after statistics competency pass.